SCL-Net: An End-to-End Supervised Contrastive Learning Network for Hyperspectral Image Classification

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***Abstract*— In recent years, deep learning (DL) presents a promising performance in hyperspectral image (HSI) classification, due to the powerful capability of automatically learning deep semantic characteristics of images. However, it is still difficult to learn highly discriminative features when limited samples are available for training a deep network. Focused on this issue, a novel end-to-end supervised contrastive learning network (SCL-Net) for spectral–spatial classification is proposed, in this article. Instead of learning features of the individual sample, the supervised contrastive learning is introduced to capture the similarity and dissimilarity distribution properties of sample pairs in a feature representation space. In this way, the need for plenty of training samples will be alleviated while an effective network training mechanism is provided for learning highly separative features. Here, SCL-Net mainly consists of one pairwise contrastive learning (PCL) subnetwork and one multilevel spectral–spatial information fusion (MLSIF) subnetwork. For the PCL subnetwork, spectral vectors are projected into deep spectral features based on convolutional operators, which are then followed by distance evaluation between "positive" pairs of similar samples and "negative" pairs of dissimilar ones. Then, a spectral distance matrix is constructed to push the network to gradually learn better features of higher intraclass compactness and interclass dispersion. For the MLSIF subnetwork, a hybrid feature-decision fusion strategy is designed, where spatial and spectral features are jointly exploited to further boost the classification performance. In specific, feature fusion is conducted by connecting low/mid/high-level spectral and spatial features via weighting, while multiple class estimations based on multilevel fusion features are adaptively integrated via probabilistic decision fusion. Overall, these two subnetworks are collaboratively trained in one framework, by optimizing a defined joint loss function consisting of a contrastive loss and a cross-entropy loss. Compared with several state-of-the-art methods, the proposed method yields a superior classification performance in terms of both objective metrics and visual performance.**

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I. Introduction

HYPERSPECTRAL imaging provides abundant spatial distribution information and spectral reflection information of land-cover objects at the same time, which greatly enhances the remote sensing Earth observation capabilities. Thus, hyperspectral image (HSI) has been widely used in many fields, e.g., precision agriculture [1], geological exploration, urban planning [2], and military defense [3], and it has become a scientific and technological frontier area developed by different countries around the world. HSI classification is used to predict the category of each pixel, which plays a core role in the application of HSIs. In recent decades, many HSI classification methods have been proposed. These methods can be summarized into two categories: spectral-based and spatial–spectral-based [4], [5].

Spectral information of HSI reflects the reflection information of objects, which can be used to identify different objects. Based on this, many spectral-based classification methods have been proposed, such as principal component analysis (PCA) [6], linear discriminant analysis (LDA) [7] and minimum noise fraction (MNF) [8]. They transform the high-dimensional spectral vectors of HSI into a low-dimensional feature space while preserving essential structural features. In addition, spectral angle mapper (SAM) [9] and spectral information divergence (SID) [10] are proposed to achieve HSI classification by means of spectral matching. The spectral-based methods have achieved good performance on HSI classification. However, single spectral information cannot effectively represent the characteristics of land-cover objects because of the phenomenon of spectral homogeneity and heterogeneity in HSIs. Besides, the flattening operation destroys the spatial structure information of HSI.

Later, many spatial–spectral-based methods have been widely studied. Benediktsson et al. [11] creatively introduced the morphological profile (MP) into HSI classification. The multiscale spatial structure features were extracted through dilation and erosion operations at different scales. Kang et al. [12] used the edge-preserving filtering technology to extract more efficient spatial–spectral features for

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classification. Recently, superpixel-based spatial–spectral classification methods have also attracted extensive attention. Superpixels are composed of a series of pixels with adjacent spatial positions and similar spectral features, which can express the irregular spatial structure information of the HSI. Lu et al. [13], [14] improved the classification results of homogeneous regions and edge regions by fusing pixel and superpixel spatial–spectral features. Fang et al. [15] combined the superpixels and sparse representation to obtain more discriminative spatial–spectral features of HSI. These approaches proved the effectiveness of the joint use of the spatial–spectral features to improve the classification performance.

In recent years, deep learning (DL) has been widely used in HSI classification via its powerful ability to adaptively extract features [16], [17], [18]. Chen et al. proposed several DL models to extract the shallow and deep spectral features, such as deep belief network (DBN) [19], stacked autoencoders (SAEs) [20], and convolutional neural network (CNN) [21]. HSIs have hundreds of bands, and therefore, 3-D convolution is a more efficient operation to jointly extract deep spatial–spectral features. Hamida et al. [22] proposed a 3-D CNN-based classification method without preprocessing and postprocessing. Furthermore, Zhong et al. [23] introduced residual structure to extract deeper features through skip connections. Recently, attention mechanisms are also widely used for hyperspectral classification. Ma et al. [24] proposed a double-branch multiattention mechanism network, which realized the saliency feature extraction of spectral–spatial information. In [25], Wang et al. introduced a multiscale spatial-channel attention mechanism and a multiscale shuffle block to improve the problem of land-cover map distortion. Feng et al. [26], [27] used the generative adversarial network (GAN) for HSI classification, and through the game process of the generator and the discriminator, the performance of the classification task is assisted and improved. Wang et al. proposed a Caps-TripleGAN framework by exploring the 1-D structure triple GAN for sample generation and integrating CapsNet for HSI classification [28]. Besides, many deep-learning-based methods have been introduced for HSI spectral–spatial classification, such as recurrent neural network (RNN) [29], graph convolutional network (GCN) [30], and transformer [31]. Compared with the traditional methods based on artificially designed features, these deep-learning-based methods can adaptively extract highly discriminative spatial–spectral features from HSI and obtain a better classification result. As a data-driven approach, DL usually requires a large amount of labeled data for training to fit the optimal model. However, it is difficult to obtain enough labeled samples. Therefore, scholars began to explore how to obtain more discriminative spatial–spectral features with limited labeled samples, so as to improve the classification accuracy [32].

More lately, contrastive learning is becoming a hot research topic in handling image classification tasks, where the key is to learn similar/dissimilar distribution properties via comparing sample pairs [33], [34]. In typical contrastive learning methods, self-supervised information is extracted to automatically generate "pseudo" labels, which are then used to learn representations of the data in the pretext tasks. As shown in

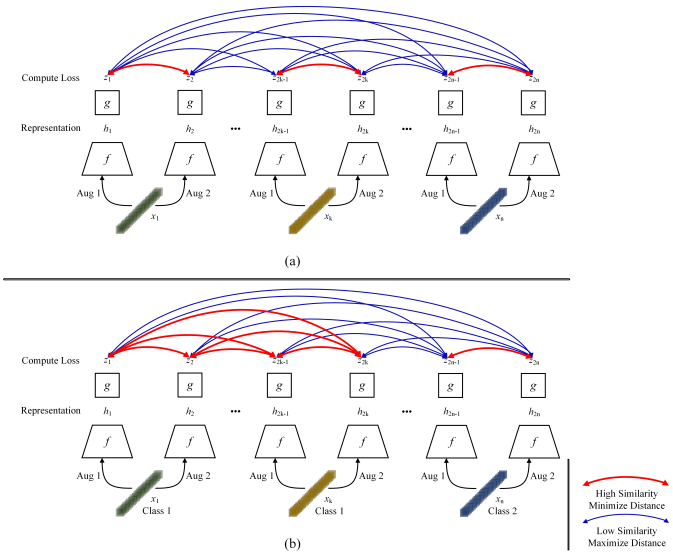

Fig. 1. (a) Self-supervised contrastive learning. (b) Supervised contrastive learning.

Fig. 1(a), for a pretext task in contrastive learning, the original image acts as an anchor, its augmented version acts as a positive sample, and the rest of the images in the batch act as negative samples. The goal is to train the network to effectively learn a feature space, where the representation of similar samples should be mapped close together, while that of dissimilar samples should be further away. Then, the learned model from the pretext task can be tuned and used for downstream tasks (e.g., classification). Due to the effectiveness of self-supervised contrastive learning, it has been exploited in the HSI classification domain. Hou et al. [35] applied the self-supervised contrastive learning model to HSI classification. The contrastive self-supervised learning (CSSL) [36] method constructed a new HSI-specific augmentation module to generate sample pairs for self-supervised contrastive learning. The contrastive learning based on transformer (CLBT) [37] combined the self-supervised contrastive learning and transformer model for classification. In addition, ContrastNet [38] combines contrastive learning and the autoencoder for hyperspectral classification. self-supervised contrastive efficient asymmetric dilated network (SC-EADNet) [39] proposed a multiscale pixelwise contextual feature learning model based on self-supervised contrastive learning. Generally, a two-state strategy of pretraining and fine-tuning is commonly adopted by these works. That is, the model is pretrained by a self-supervised contrastive learning subnetwork without using any label information, and then the feature embedding parameters of the pretrained model are tuned by training a supervised classification network with the exploit of label information. However, data augmentation plays a key role in these methods to enhance performance, which actually relies on enough professional knowledge and careful design. In addition, as shown in Fig. 1(a), the way of obtaining "negative pairs" in the self-supervised contrastive learning frameworks may lead to samples belonging to the same class being far in the feature space [40]. In contrast, supervised contrastive learning can make full use of limited labeled training samples, which can help enhance the

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separability between different classes and class aggregation among the same classes, as shown in Fig. 1(b). Moreover, it is not easy to obtain a huge amount of data, which is necessarily required for learning highly representative features in the pretraining stage. In this case, it is difficult to use the quite limited training samples to effectively fine-tune the network in the fine-tuning stage, and thus, the final classification performance is limited. In contrast, we propose an end-to-end supervised contrastive learning network (SCL-Net), instead of the "pretrain and fine-tune" network, which can jointly use the good properties of contrastive learning and training sample information in enhancing the discriminative representation learning capability.

Based on the above analysis, we propose an end-to-end SCL-Net for HSI spectral–spatial classification by effectively leveraging label information, in this article. First, the supervised pairwise contrastive learning (PCL) subnetwork is designed to obtain discriminative spectral feature representations, while the convolution-based subnetwork is exploited to extract spatial features. Then, the multilevel spectral and spatial features are integrated with a hybrid fusion scheme, where both the weighting feature fusion and adaptive probabilistic decision fusion strategies are designed in one framework. By collaboratively training both the subnetworks, more accurate classification results can be obtained. The main contributions are concluded as follows.

1) In this work, a novel end-to-end contrastive learning network is introduced for spectral–spatial classification of HSI. Different from the traditional DL network where features are learned from individual labeled samples, this work provides an effective scheme to learn from similarity and dissimilarity properties of sample pairs. This can help the network effectively learn a feature representation space, where "positive" sample pairs will be pulled together while "negative" sample pairs will depart far away.
2) To alleviate the need for training samples for discriminative feature representation learning, this method provides one supervised contrastive learning strategy. This is achieved by comparing each pair of labeled samples and constructing a distance matrix to guide the optimization of the feature learning network. Compared with the original few training samples, comparison between sample pairs can help increase the amount of learned information, e.g., n samples corresponds to C_n^2 sample pairs. In this way, the network can effectively enhance the learning ability of discriminative information with limited labeled samples.
3) A hybrid spectral–spatial information fusion scheme is designed to further enhance the classification performance. On one hand, spectral–spatial fusion features are generated via weighting feature fusion, which can help increase the discriminative capability. On the other hand, low/mid/high-level spectral-spatial fusion features are used for classification estimations, which are then adaptively fused to improve the robustness of classification.

The remainder of this article is organized as follows. Section II introduces the details of the proposed method. The experimental results and analysis of the PaviaU, Houston, and Kennedy Space Center (KSC) datasets are provided in Section III. In the end, Section IV summarizes our work.

II. Proposed Method

The flowchart of the proposed method SCL-Net is shown in Fig. 2; the SCL-Net mainly consists of two modules: PCL and multilevel spatial–spectral information fusion (MLSIF). First, the spectral pixels are input to the PCL module to extract the multilevel spectral features SF^1, SF^2, and SF^3. During this step, spectral contrastive learning is designed to extract spectral features of intraclass compactness and interclass dispersion. Second, the spatial patches are input to the MLSIF module and the discriminative spectral features guided the process of fusion to obtain the multilevel spectral–spatial features F^1, F^2, and F^3. Finally, the adaptive probability fusion strategy is introduced to obtain the final class probability P. The detailed descriptions are given below.

A. Pairwise Contrastive Learning

The spectral pixels of HSIs contain the reflection information of the land-cover objects, which can be used to identify different objects. Besides, an efficient feature space for classification should have two elements: accurately identify a single object and describe the characteristics of intraclass compactness and interclass dispersion of objects. Based on those considerations, we construct the PCL module. PCL consists of two steps: spectral feature extraction and spectral contrastive learning. From the network structure, the PCL branch consists of three sequentially connected spectral feature extraction blocks, and each block consists of a fully connected layer, batch normalization layer, and ReLu layer connected in sequence. The details are as follows.

The multilevel features can describe both deep semantic information and shallow appearance information. Thus, the multilevel spectral features are extracted at first. This process can be expressed mathematically as follows:

$$SF_i^k = \begin{cases} \sigma(w * x_i), & k = 1 \\ \sigma\left(w * SF_i^{k-1}\right), & k = 2, 3. \end{cases} \tag{1}$$

Here, (1) represents the extraction process of spectral features, x_i is the center pixel of the ith image patch O_i, w is the full connection layer, σ is the nonlinear activation function, and SF_i^k means the kth spectral feature vector from x_i.

Furthermore, spectral contrastive learning is designed to extract the discriminative spectral features. The implementation details are described below. The positive and negative sample pairs are constructed according to the label in a minibatch. As shown in (2), if the two samples are from different classes, the pairwise label A_{ij} is set to be 0, and if the two samples are from the same class, the pairwise label A_{ij} is set to 1

$$A_{ij} = \begin{cases} 0, & y_i \neq y_j \\ 1, & y_i = y_j. \end{cases} \tag{2}$$

Here, y_i is the label of the input x_i.

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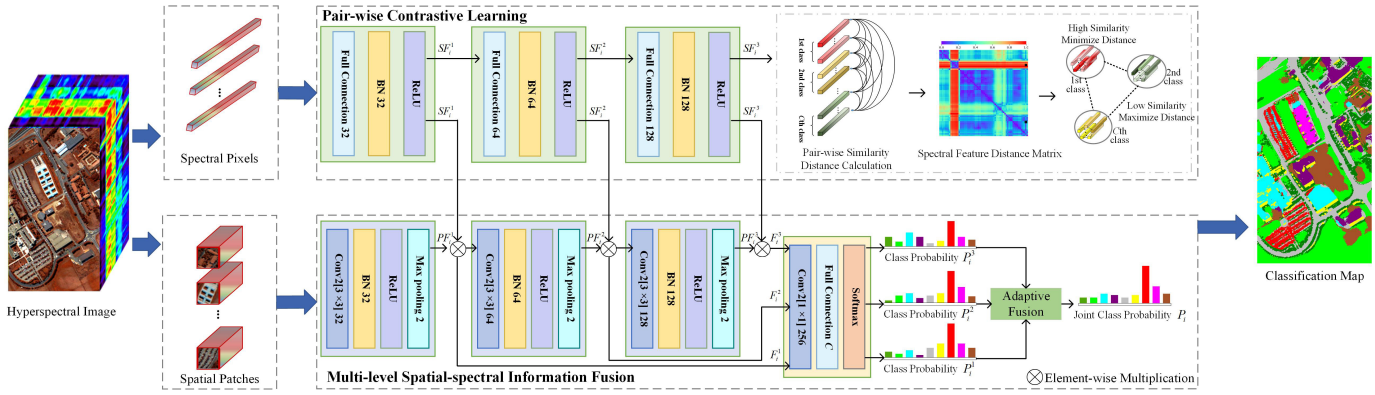

Fig. 2. Flowchart of the proposed method SCL-Net.

Then, we construct a spectral feature distance matrix to measure the similarity between pairwise samples in a mini-batch. The similarity distance D is calculated according to the Euclidean distance. The greater the distance, the smaller the similarity. It can be described as follows:

$$D\left(SF_i^3, SF_j^3\right) = \left|\left|SF_i^3 - SF_j^3\right|\right|_2^2. \tag{3}$$

In this course, the contrastive loss is used to supervise the optimization of the spectral feature distance matrix. The contrastive loss can be expressed as

$$\mathcal{L}_1 = \frac{1}{2N_1}\sum_{n_{ij}=1}^{N_1}\left(A_{ij}D^2 + \left(1 - A_{ij}\right)\max(0, \zeta - D)^2\right) \tag{4}$$

where N_1 is the number of sample pairs, and ζ is a hyperparameter which is empirically set to 2. Each pairwise label A_{ij} is calculated based on the label of the n_{ij}th sample pair, i.e., x_i and x_j.

For the positive samples, the pairwise A_{ij} is set to 1, and $\mathcal{L}_1$ is only calculated based on the similarity of positive samples. By minimizing the value of $\mathcal{L}_1$, the features with intraclass compactness are obtained. For the negative samples, the pairwise A_{ij} is set to 0. Because DL is fitting the optimal model by minimizing the loss function, the hyperparameter ζ is used to maximize the distance between negative samples. Meanwhile, the features with interclass dispersion are obtained by minimizing the value of $\mathcal{L}_1$.

The spectral feature distance matrix is shown in Fig. 2, which is a diagonal matrix, and the diagonal line of the matrix represents the spectral similarity between same samples. After the training process, samples from the same class will share a small feature distance, that is, the similarity of the embedding is high, and vice versa. In addition, the pairwise strategy increases the learning capability of detailed information with limited labeled samples. For the n labeled samples, the number of pairwise samples can be calculated using the combination, which is equal to C_n^2. In this way, the amount of learning information is greatly increased with limited labeled samples. Moreover, the supervision information includes not only the category information of single sample but also the correlation information between samples. In summary, the discriminative spectral features are obtained.

B. Multilevel Spatial–Spectral Information Fusion

Many research works have demonstrated that the joint use of spectral–spatial information can greatly improve the performance of HSI classification. In this section, based on the extraction of discriminative spectral features, an MLSIF framework is constructed. Specifically, MLSIF performs information fusion from two aspects: spectral–spatial feature fusion and multilevel decision fusion. According the network structure, the MLSIF branch consists of three sequentially connected spatial feature extraction blocks, and each block consists of a convolutional layer, batch normalization layer, ReLu layer, and max pooling layer connected in sequence. Then the fused spatial–spectral features are passed through three classifiers to obtain the initial classification results. The structure of the classifier is sequentially connected convolutional layer, fully connected layer, and softmax. The details are as follows.

First, the multilevel spatial features are extracted by convolution operation, which can be formulated as follows:

$$PF_i^k = \begin{cases} \text{Max pool}(\sigma\left(W * O_i + b\right)), & k = 1 \\ \text{Max pool}\left(\sigma\left(W * PF_i^{k-1} + b\right)\right), & k = 2, 3 \end{cases} \tag{5}$$

where O_i is the ith input image patch, W is the convolution kernel parameter, b is the bias, and PF_i^k is the kth spatial feature matrix from O_i.

Different from simply concatenating or adding features, the spectralwise multiplication operation is constructed to fuse the spectral and spatial features. Here, the discriminative spectral features SF_i^k of size $1 \times 1 \times B$ and the corresponding level spatial features PF_i^k of size $W \times W \times B$ are multiplied along the spectral dimension. In this way, similar to the attention mechanism, the channels with more discriminative spectral–spatial information have been emphasized. The spectral–spatial features of size $W \times W \times B$ are obtained. The mathematical formula is expressed as follows:

$$F_i^k = f\left(SF_i^k, PF_i^k\right) \tag{6}$$

where $f(\cdot)$ represents the operation of multiplying each element in the vector SF_i^k with each feature matrix in PF_i^k.

To further improve the performance, the adaptive fusion strategy (AFS) is introduced to predict the final classification map. The low-, mid-, and high-level spatial–spectral features

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passed through three classifiers, respectively, to obtain the initial class probabilities P_i^1, P_i^2, and P_i^3. Finally, the initial class probabilities P_i^1, P_i^2, and P_i^3 are combined to predict the final probability P_i, and then the predicted probability p_i corresponding to x_i is obtained by the softmax function, which can be described as follows:

$$p_i = \text{Softmax}(P_i) \quad \text{s.t. } P_i = \gamma_1 P_i^1 + \gamma_2 P_i^2 + \gamma_3 P_i^3. \tag{7}$$

Here, the learnable parameters γ_1, γ_2, and γ_3 are adaptively learned to obtain the optimal values in the iterative process, which avoids the inaccuracy of manual design. Concretely, the values of the parameters are continuously adjusted by backpropagation until the network converges.

In this stage, the cross-entropy loss function is used to supervise the classification process, which can be described as

$$\mathcal{L}_2 = -\frac{1}{N_2}\sum_{i=1}^{N_2}\sum_{j=1}^{C}\left(y_i^j \log\left(p_i^j\right)\right). \tag{8}$$

Here, N_2 is the number of samples and C is the total number of land-cover classes. For y_i^j, when the label of x_i is equal to j, y_i^j is set to 1; otherwise, y_i^j is set to 0. p_i^j is the probability that x_i is predicted to be jth class

$$\mathcal{L} = \mathcal{L}_2 + \lambda \mathcal{L}_1. \tag{9}$$

In summary, the cross-entropy loss function $\mathcal{L}_2$ and the contrastive loss function $\mathcal{L}_1$ are used jointly during the training process. Besides, the two items in (9) are balanced by a predefined weight λ. Along with the optimization of the loss function, the final classification map will have higher accuracy.

III. Experimental Results and Discussion

A. Data Description

In this section, three popular hyperspectral datasets are used to evaluate the performance of the proposed method: Pavia University, Houston, and KSC.

1) PaviaU: The PaviaU dataset captures the area involving the University of Pavia in Italy by the reflective optics system imaging spectrometer (ROSIS) sensor. This HSI consists of 115 bands with 610 × 340 spatial size, and the spectral range is from 0.43 to 0.86 μm. The spatial resolution of the image is 1.3 m. Among them, 12 bands are removed due to the influence of noise. There are nine classes of objects included in the image, and the number of labeled samples is 42 766. The false-color image and ground truth are shown in Fig. 3.

2) Houston: The Houston dataset was collected using the ITRES CASI-1500 sensor over the University of Houston in TX, USA. The image size is 349 × 1905 pixels with 144 spectral bands ranging from 0.364 to 1.046 μm. It should be noted that the used dataset is a cloud-free HS product. There are 15 classes of objects included in the image, and the number of labeled samples is 11 099. In Fig. 4, we show a false-color image and the ground truth of the Houston dataset.

3) KSC: The KSC dataset was collected by the airborne visible infrared imaging spectrometer (AVIRIS) instrument over the KSC, Merritt Island, FL,

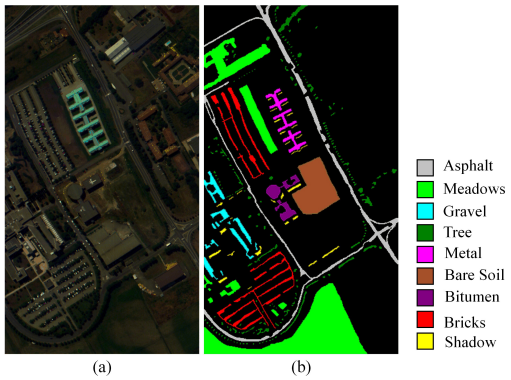

Fig. 3. PaviaU dataset. (a) False color image. (b) Ground truth.

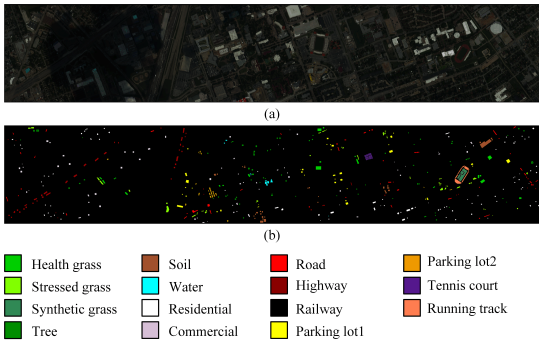

Fig. 4. Houston dataset. (a) False color image. (b) Ground truth.

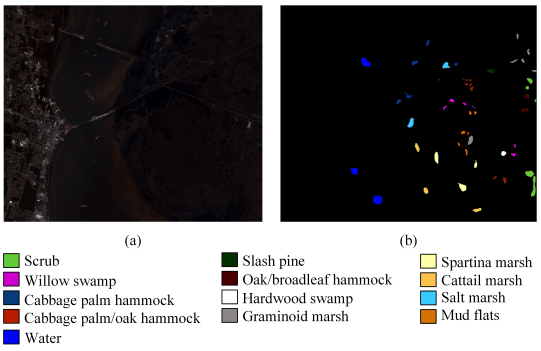

Fig. 5. KSC dataset. (a) False color image. (b) Ground truth.

USA. Once noisy bands have been removed, this dataset contains 176 bands with 512 × 614 spatial sizes, ranging from 0.4 to 2.5 μm and with a spatial resolution of 20 m. A total of 5211 pixels labeled in 13 classes are considered for classification purposes. The false-color image and ground truth are shown in Fig. 5.

B. Experimental Setting

In our experiments, the proposed method SCL-Net is implemented on a single NVIDIA GeForce GTX 1080ti GPU with PyTorch. The Adam optimizer is used to train this network. The learning rate is set to be 1e-3, and the training epochs are 400. Tables I–III report the number of training samples and test samples, as well as the total number in each class on three datasets. The data values are normalized to the

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TABLE I
SAMPLE DISTRIBUTIONS ON THE PAVIAU DATASET

Class NO.	Class Name	Train	Test	Total
1	Asphalt	10	6621	6631
2	Meadows	10	18639	18649
3	Gravel	10	2089	2099
4	Tree	10	3054	3064
5	Meatal	10	1335	1345
6	Bare Soil	10	5019	5029
7	Bitumen	10	1320	1330
8	Bricks	10	3672	3682
9	Shadows	10	937	947
	Total	90	42686	42776

TABLE II
SAMPLE DISTRIBUTIONS ON THE HOUSTON DATASET

Class No.	Class Name	Train	Test	Total
1	Healthy grass	10	1063	1073
2	Stressed grass	10	800	810
3	Synthetic grass	10	687	697
4	Tree	10	1043	1053
5	Soil	10	1232	1242
6	Water	10	315	325
7	Residential	10	968	978
8	Commerucial	10	614	624
9	Road	10	1021	1031
10	Highway	10	372	382
11	Railway	10	104	114
12	Parking Lot1	10	1223	1233
13	Parking Lot2	10	439	449
14	Tennis court	10	418	428
15	Running track	10	650	660
	Total	150	10949	11099

TABLE III
SAMPLE DISTRIBUTIONS ON THE KSC DATASET

Class NO.	Class Name	Train	Test	Total
1	Scrub	10	751	761
2	Willow swamp	10	233	243
3	Cabbage palm hammock	10	246	256
4	Cabbage palm/oak hammock	10	242	252
5	Slash pine	10	151	161
6	Oak/broakleaf hammock	10	219	229
7	Hardwood swamp	10	95	105
8	Graminoid	10	421	431
9	Spaitina marsh	10	510	520
10	Cattail marsh	10	394	404
11	Salt marsh	10	409	419
12	Mud flats	10	493	503
13	Water	10	917	927
	Total	130	5081	5211

range [0, 1] along the spectral dimension. Most classification methods use data augmentation to improve the robustness of the model. The purpose of the proposed method to use data augmentation is expanding the amount of learnable samples, preventing the overfitting and enhancing the robustness of the network. Following them, the flipping and random noise enhancement are used on the input HSI images to generate augmented data, which are also adopted for all the compared DL-based methods for a fair comparison. In the proposed method, random augmentation is used at different epochs, and its augmented samples are not compared with themselves in the same epoch.

To measure the performance of the proposed method, four objective metrics are used to evaluate the classification accuracies, i.e., class accuracy (CA), overall accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa). The parameter λ in the loss function is set to be 0.5. The spatial size of the input HSI cube is 24×24. In addition, the parameters of the methods in the comparison experiment are set as the default parameters in the original articles. To avoid the contingency of the experiments, the training samples are selected randomly, and the experiments are repeated ten times to obtain the mean and standard variances of CA, OA, AA, and Kappa.

C. Parameter Discussion

For the proposed method SCL-Net, there are two important parameters that need to be predefined: the spatial size N of the input data and the parameter λ in the loss function.

1) Spatial Size N of Image Patch: Experiments were designed with a grid search to find the most suitable spatial size N, $N \in \{8, 16, 24, 32\}$. Fig. 6 shows the influence of N on the classification performance, in terms of three objective metrics, i.e., OA, AA, and Kappa. It can be seen that along with the increase in N, the values of OA, AA, and Kappa all show a trend of first increasing and then decreasing. In specific, the highest values for these three metrics can be obtained when $N = 24$. Because HSI has continuity in space, the smaller spatial size will lose much useful spatial information. On the contrary, the method uses the label of the center pixel representing the label of the entire image block, and larger spatial size will introduce irrelevant information. Based on the above analysis, the value of N is set to 24 in the proposed method.

2) Parameter λ in the Loss Function: Experiments were designed with a grid search to find the most suitable parameter λ, $\lambda \in \{0.25, 0.50, 0.75, 1.00\}$. Fig. 7 shows the influence of λ on the classification performance, in terms of three objective metrics, i.e., OA, AA, and Kappa. It can be seen that the highest values for these three metrics can be obtained when $\lambda = 0.50$ on the PaviaU, Houston, and KSC datasets. If the weight of λ is too small, the constraints on the spectral features are weak, and conversely, the constraints on the spectral features are too strong, and the spectral features of intraclass compactness and interclass discrete characteristics cannot be obtained. To unify the experimental settings, the value of λ is set to 0.50 in the proposed method.

D. Ablation Studies

The double-branch fusion framework (DBFF), the contrastive learning module (CLM), and the AFS are essential components of SCL-Net. To validate their effectiveness of them, we designed ablation experiments on all the three datasets. The results are shown in Table IV, and the best results in this table have been marked in bold.

In specific, to study the impact of DBFF on the proposed network, we designed a new method that removes the pixel branch in SCL-Net. The method is named by w/o (i.e., without) DBFF, and the result is shown in the first column of Table IV. It can be observed that SCL-Net achieves higher

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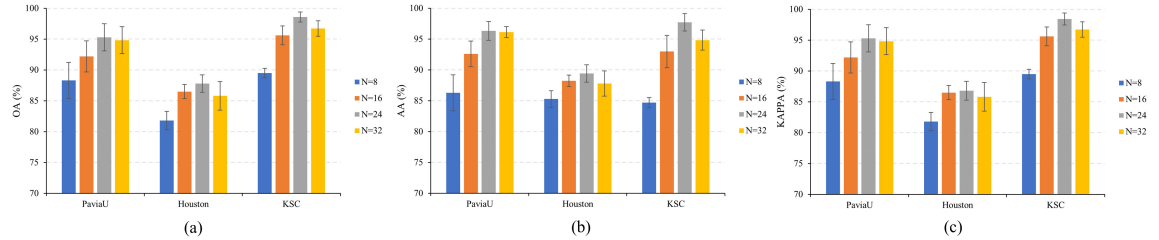

Fig. 6. Influence of the parameter N of image patch size on the classification performance of the proposed method in terms of (a) OA, (b) AA, and (c) Kappa.

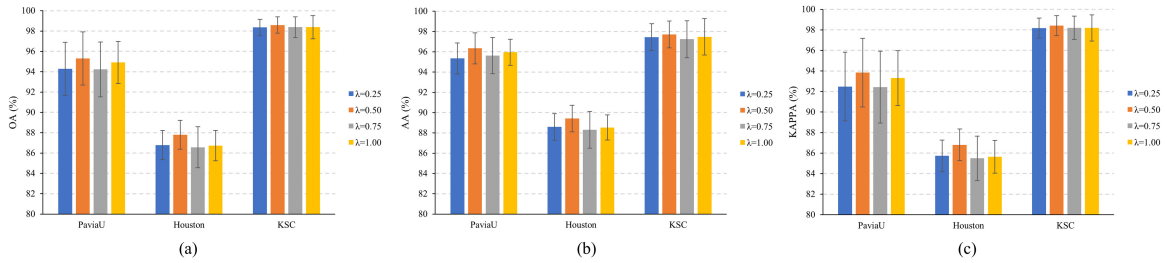

Fig. 7. Influence of the parameter λ in the loss function on the classification performance of the proposed method in terms of (a) OA, (b) AA, and (c) Kappa.

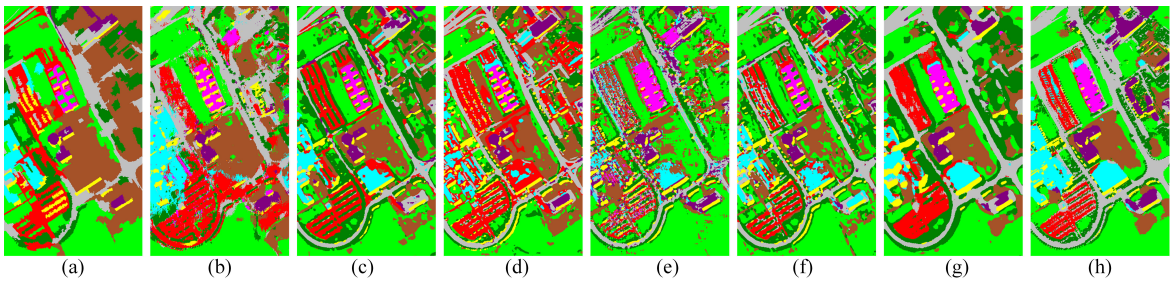

Fig. 8. Classification maps for the PaviaU dataset. (a) SBD. (b) PCA-EPF. (c) SPRN. (d) SSRN. (e) RSSAN. (f) A2S2K. (g) MTGAN. (h) SCL-Net.

TABLE IV
COMPARISON OF QUANTITATIVE INDEXES OF ABLATION STUDIES

Dataset	Metrics	w/o DBFF	w/o CLM	w/o AFS	SCL-Net
PaviaU	OA	90.7 ± 1.92	94.24 ± 2.31	94.08 ± 2.13	$\mathbf{95.30 \pm 2.22}$
	AA	89.23 ± 1.31	96.20 ± 1.33	94.64 ± 1.70	$\mathbf{96.34 \pm 1.53}$
	Kappa	87.79 ± 2.44	92.49 ± 2.94	92.20 ± 2.85	$\mathbf{93.84 \pm 2.87}$
Houston	OA	83.60 ± 1.44	86.89 ± 1.32	87.51 ± 1.40	$\mathbf{87.80 \pm 1.40}$
	AA	85.73 ± 1.49	88.56 ± 1.19	89.27 ± 1.25	$\mathbf{89.42 \pm 1.42}$
	Kappa	82.28 ± 1.55	85.83 ± 1.43	86.50 ± 1.51	$\mathbf{86.81 \pm 1.52}$
KSC	OA	97.74 ± 0.39	97.34 ± 0.91	98.32 ± 0.61	$\mathbf{98.59 \pm 0.87}$
	AA	96.46 ± 0.77	95.76 ± 1.31	97.36 ± 0.90	$\mathbf{97.71 \pm 1.41}$
	Kappa	97.47 ± 0.43	97.03 ± 1.02	98.14 ± 0.68	$\mathbf{98.42 \pm 0.97}$

values on three comprehensive evaluation metrics of OA, AA, and Kappa. Specifically, the OA of SCL-Net is 4.60%, 4.20%, and 0.85% higher than that without DBFF on three datasets, which demonstrates the effectiveness of DBFF.

To validate the effectiveness of CLM, we remove the contrastive loss in the loss function and only the cross entropy function is used to train the whole network, named by w/o (i.e., without) CLM. The results are shown in the second column of Table IV. Compared with SCL-Net, the three metrics by the w/o CLM method have different degrees of decrease on all the three datasets. Numerically, SCL-Net is 1.06%, 0.91%, and 1.25% higher in OA than that without CLM. The comparison results demonstrate that the proposed SCL-Net has a better performance.

Besides, to validate the effectiveness of AFS, we only use the high-level spatial–spectral features to predict the final class probability, named by w/o (i.e., without) AFS. The results are shown in the third column of Table IV. Compared with SCL-Net, the three metrics by the w/o AFS method are significantly reduced. In contrast, SCL-Net takes advantage of the complementary information by adaptive fusing the low-, mid-, and high-level features. Meanwhile, the weights γ_1, γ_2, and γ_3 are obtained through adaptive learning, which avoids the inaccuracy of manual design.

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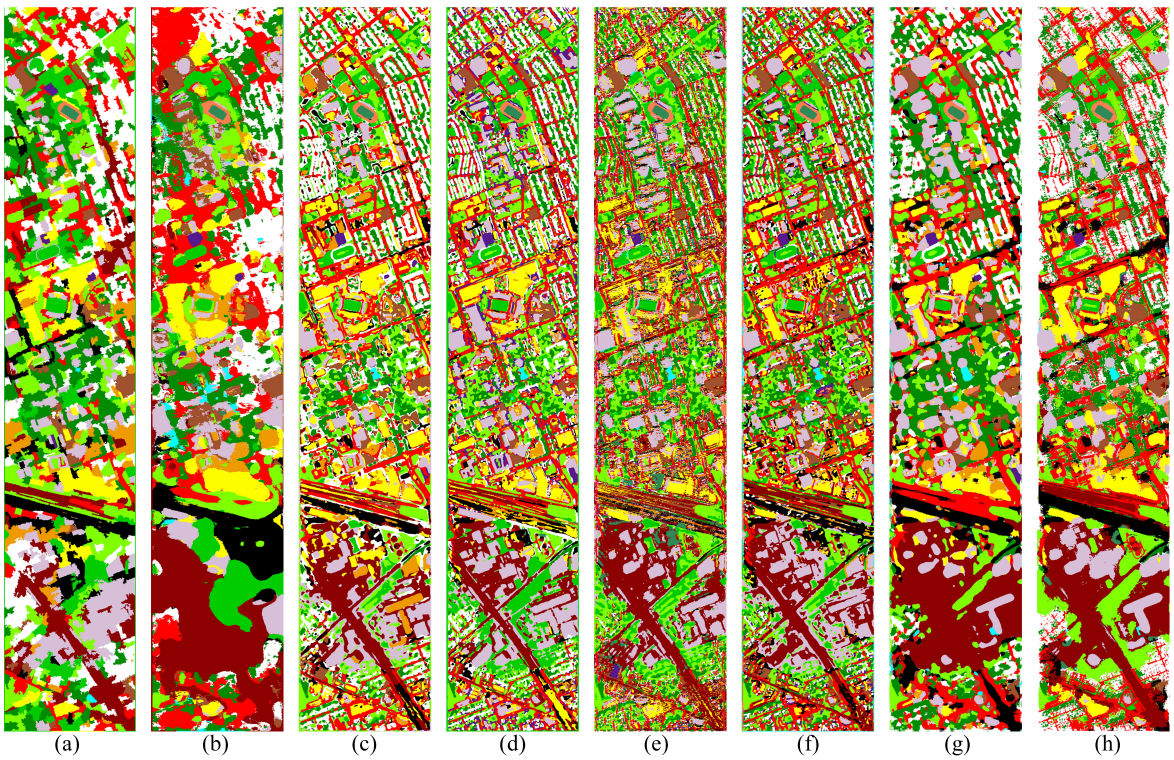

Fig. 9. Classification maps for the Houston dataset. (a) SBD. (b) PCA-EPF. (c) SPRN. (d) SSRN. (e) RSSAN. (f) A2S2K. (g) MTGAN. (h) SCL-Net.

TABLE V
CLASSIFICATION RESULTS FOR THE PAVIAU DATASET USING TEN PER CLASS OF THE LABELED DATA FOR TRAINING

Class	SBD [41]	PCA-EPF [12]	SPRN [42]	SSRN [23]	RSSAN [43]	A2S2K [44]	MTGAN [27]	SCL-Net
1	70.87 ± 6.95	77.27 ± 15.04	83.03 ± 8.05	67.06 ± 21.44	77.54 ± 0.09	88.58 ± 5.68	89.63 ± 5.65	$\mathbf{94.23 \pm 4.7}$
2	88.62 ± 8.88	$\mathbf{98.46 \pm 1.00}$	91.17 ± 6.92	82.31 ± 15.20	91.94 ± 3.48	92.83 ± 4.28	93.34 ± 6.69	94.65 ± 4.92
3	$\mathbf{93.91 \pm 7.72}$	77.52 ± 6.78	76.00 ± 11.36	64.83 ± 20.76	56.76 ± 8.72	86.35 ± 8.60	84.54 ± 12.49	92.08 ± 8.42
4	55.50 ± 4.28	89.70 ± 4.26	91.58 ± 3.81	91.80 ± 4.01	89.67 ± 3.01	95.18 ± 3.04	92.74 ± 4.59	$\mathbf{96.33 \pm 1.12}$
5	94.55 ± 1.16	88.25 ± 12.68	$\mathbf{99.87 \pm 0.18}$	99.69 ± 0.62	97.65 ± 2.81	99.67 ± 0.60	98.98 ± 1.35	99.85 ± 0.15
6	95.95 ± 4.76	90.37 ± 6.04	71.99 ± 15.62	58.89 ± 26.27	68.25 ± 10.44	65.05 ± 11.80	92.58 ± 3.41	$\mathbf{96.77 \pm 4.60}$
7	92.46 ± 2.51	71.49 ± 13.59	97.74 ± 2.52	90.09 ± 22.93	78.71 ± 12.25	93.50 ± 4.52	94.30 ± 7.37	$\mathbf{97.80 \pm 3.17}$
8	88.61 ± 10.03	59.63 ± 8.74	$\mathbf{96.04 \pm 2.91}$	87.75 ± 7.92	65.06 ± 13.94	90.07 ± 6.17	89.66 ± 8.59	95.92 ± 3.26
9	79.53 ± 3.43	76.79 ± 12.92	99.47 ± 0.43	$\mathbf{99.88 \pm 0.19}$	95.55 ± 4.96	99.71 ± 0.36	93.34 ± 4.62	99.44 ± 0.98
OA	84.72 ± 3.80	84.56 ± 6.02	88.01 ± 2.60	78.65 ± 7.30	82.57 ± 2.07	88.90 ± 2.58	92.09 ± 2.98	$\mathbf{95.30 \pm 2.22}$
AA	84.44 ± 1.83	81.05 ± 4.22	89.65 ± 1.18	82.48 ± 4.90	80.12 ± 3.0	90.11 ± 1.87	92.12 ± 1.66	$\mathbf{96.34 \pm 1.53}$
Kappa	80.14 ± 4.52	80.29 ± 7.00	84.24 ± 3.26	72.52 ± 8.01	76.92 ± 2.71	85.31 ± 3.38	89.65 ± 3.76	$\mathbf{93.84 \pm 2.87}$

E. Performance Comparison

In this section, the performance of the proposed method is compared with five recently proposed DL-based spatial–spectral classification methods, including spectral partitioning residual network (SPRN) [42], spectral–spatial residual network (SSRN) [23], residual spectral–spatial attention network (RSSAN) [43], A2S2K [44], and multitask generative adversarial network (MTGAN) [27]. The SPRN splits the input spectral bands into several nonoverlapping continuous subbands and uses cascaded parallel improved residual blocks to extract spectral–spatial features from these subbands. The SSRN also uses DBFF, which adopts two residual connections blocks to learn spectral and spatial representations separately. The RSSAN embeds a sequential spectral–spatial attention module in a residual block for the adaptive learning of spatial–spectral information. The A2S2K combines spectral attention with improved 3-D ResBlocks and adopts an efficient feature recalibration mechanism to boost the classification performance. The MTGAN is a multitask GAN, which simultaneously undertakes the reconstruction task and the classification task to improve the discrimination and generalization ability of the classification task. And MTGAN is aimed at how to improve the classification performance with limited labeled samples. To further demonstrate the superiority of the proposed method, we compare with two standard (non-DL-based) comparison methods, those integrating textural information or using the dimensionality reduction as a preprocessing step. superpixel-based Brownian descriptor (SBD) [41] used superpixel segmentation to extract shape-adaptive spatial structure information from dimension-reduced HSI. Then HSI image blocks are jointly represented by local spectral–spatial features based on the Brownian descriptor (BD). PCA-edge preserving features (EPF) [12] constructs the standard EPFs by applying edge-preserving filters, and then the spectral dimension of the stacked EPFs is reduced with PCA.

Table V reports the OA, AA, Kappa, and the classification accuracies of all the classes on the PaviaU dataset. For each line, the bold fonts indicate the best performance.

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TABLE VI
CLASSIFICATION RESULTS FOR THE HOUSTON DATASET USING TEN PER CLASS OF THE LABELED DATA FOR TRAINING

Class	SBD [41]	PCA-EPF [12]	SPRN [42]	SSRN [23]	RSSAN [43]	A2S2K [44]	MTGAN [27]	SCL-Net
1	82.22 ± 4.62	80.14 ± 9.23	**93.25 ± 6.08**	91.00 ± 6.77	86.45 ± 6.93	92.49 ± 6.55	82.47 ± 4.29	83.68 ± 8.04
2	78.31 ± 9.23	74.69 ± 10.30	**96.31 ± 4.71**	93.47 ± 5.31	85.03 ± 7.11	94.17 ± 5.04	92.88 ± 5.01	90.42 ± 6.70
3	**100.00 ± 0.00**	**100.00 ± 0.00**	99.71 ± 0.52	97.15 ± 6.76	99.46 ± 0.98	99.70 ± 0.47	98.55 ± 0.95	99.96 ± 0.09
4	75.38 ± 1.85	57.04 ± 7.04	91.65 ± 2.83	**93.75 ± 3.34**	87.57 ± 4.80	92.50 ± 3.50	89.94 ± 3.53	92.23 ± 1.63
5	**99.74 ± 0.36**	96.64 ± 3.64	99.69 ± 0.44	97.53 ± 3.63	93.66 ± 3.98	99.65 ± 0.38	98.21 ± 1.66	97.49 ± 2.78
6	40.32 ± 18.98	88.44 ± 6.90	**93.56 ± 5.01**	86.26 ± 10.99	81.84 ± 6.27	91.37 ± 6.92	89.05 ± 7.60	92.03 ± 6.94
7	64.50 ± 7.00	63.44 ± 6.75	68.67 ± 14.96	73.08 ± 16.62	68.14 ± 7.58	74.83 ± 10.13	85.22 ± 5.38	**90.42 ± 4.14**
8	56.16 ± 4.47	**72.82 ± 10.92**	59.37 ± 7.06	61.02 ± 8.49	56.35 ± 9.71	64.93 ± 6.91	62.21 ± 8.42	63.20 ± 6.20
9	70.47 ± 7.93	47.74 ± 4.67	78.22 ± 6.56	71.61 ± 7.65	71.85 ± 6.90	71.79 ± 4.37	84.32 ± 3.39	**88.66 ± 6.05**
10	**89.79 ± 9.46**	72.34 ± 9.62	81.93 ± 11.29	47.81 ± 18.97	73.30 ± 16.86	76.24 ± 11.13	68.95 ± 17.51	83.34 ± 13.65
11	75.02 ± 5.88	**89.92 ± 4.84**	75.10 ± 7.34	69.80 ± 9.01	54.09 ± 10.35	76.12 ± 9.14	83.54 ± 8.25	82.48 ± 8.66
12	69.53 ± 14.57	76.38 ± 5.23	83.72 ± 4.73	80.11 ± 11.38	75.01 ± 12.85	**88.45 ± 4.41**	86.92 ± 8.02	86.33 ± 5.58
13	75.73 ± 5.22	63.28 ± 16.26	73.94 ± 9.98	87.56 ± 7.68	67.19 ± 10.79	**95.21 ± 0.74**	94.73 ± 2.01	91.46 ± 3.52
14	72.06 ± 0.32	96.11 ± 2.06	99.93 ± 0.15	99.83 ± 0.36	92.82 ± 4.20	99.83 ± 0.37	99.14 ± 0.93	**99.71 ± 0.58**
15	100.00 ± 0.00	99.22 ± 0.61	99.63 ± 0.74	99.91 ± 0.20	92.94 ± 6.13	99.88 ± 0.15	99.11 ± 0.98	**99.92 ± 0.10**
OA	77.35 ± 1.73	74.35 ± 1.64	84.48 ± 2.26	81.04 ± 2.41	77.72 ± 1.88	85.63 ± 1.61	85.77 ± 1.37	**87.80 ± 1.40**
AA	76.62 ± 1.54	78.40 ± 1.46	86.11 ± 1.92	83.76 ± 2.33	79.27 ± 1.97	87.81 ± 1.42	87.68 ± 1.24	**89.42 ± 1.42**
Kappa	75.50 ± 1.86	72.28 ± 1.76	83.22 ± 2.45	80.20 ± 1.67	75.93 ± 2.03	84.47 ± 1.74	84.62 ± 1.48	**86.81 ± 1.52**

TABLE VII
CLASSIFICATION RESULTS FOR THE KSC DATASET USING TEN PER CLASS OF THE LABELED DATA FOR TRAINING

Class	SBD [41]	PCA-EPF [12]	SPRN [42]	SSRN [23]	RSSAN [43]	A2S2K [44]	MTGAN [27]	SCL-Net
1	97.70 ± 2.28	99.52 ± 1.06	98.51 ± 1.48	98.92 ± 0.90	98.82 ± 1.05	98.05 ± 0.72	99.73 ± 0.45	**99.82 ± 0.21**
2	69.57 ± 10.17	99.83 ± 0.30	92.85 ± 4.89	82.17 ± 9.66	75.92 ± 17.32	82.50 ± 10.52	92.51 ± 11.61	**96.23 ± 7.64**
3	78.01 ± 6.58	99.50 ± 1.59	95.53 ± 6.75	85.23 ± 16.52	89.88 ± 6.81	86.90 ± 13.42	94.13 ± 12.11	**99.50 ± 0.74**
4	85.54 ± 7.32	**96.14 ± 4.42**	51.24 ± 13.01	32.44 ± 19.38	39.81 ± 7.20	30.03 ± 17.62	75.04 ± 20.24	88.29 ± 14.14
5	88.61 ± 8.24	89.51 ± 12.57	76.01 ± 10.44	61.15 ± 27.62	57.17 ± 14.29	45.55 ± 37.79	65.78 ± 20.11	**90.73 ± 8.47**
6	90.87 ± 6.66	85.20 ± 6.77	92.95 ± 5.58	83.97 ± 14.00	79.10 ± 7.57	78.79 ± 17.64	92.49 ± 6.45	**99.19 ± 1.71**
7	96.63 ± 4.13	94.50 ± 13.91	96.14 ± 3.94	92.87 ± 12.79	98.25 ± 3.24	91.34 ± 22.89	98.25 ± 2.04	**99.88 ± 0.33**
8	90.95 ± 10.78	96.13 ± 7.11	97.36 ± 2.06	81.84 ± 19.60	75.19 ± 13.53	84.88 ± 11.78	99.52 ± 1.12	**99.55 ± 0.63**
9	94.43 ± 6.89	**99.94 ± 0.13**	97.52 ± 3.29	94.03 ± 10.30	85.27 ± 28.44	94.34 ± 5.70	96.43 ± 5.55	99.37 ± 1.20
10	96.65 ± 3.00	97.59 ± 3.80	96.87 ± 2.30	94.22 ± 4.92	83.98 ± 5.61	95.68 ± 2.12	98.53 ± 1.70	**99.49 ± 0.78**
11	97.60 ± 0.10	94.90 ± 10.76	98.21 ± 2.17	99.24 ± 1.11	98.86 ± 1.32	99.35 ± 1.53	99.02 ± 1.54	**99.97 ± 0.08**
12	86.31 ± 3.18	98.09 ± 3.30	95.76 ± 4.27	93.15 ± 4.59	75.28 ± 7.87	90.22 ± 7.83	98.02 ± 3.55	**98.15 ± 1.75**
13	**100.00 ± 0.00**	94.50 ± 13.91	**100.00 ± 0.00**	99.85 ± 0.44	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**
OA	92.63 ± 0.01	97.01 ± 2.60	94.56 ± 0.83	89.80 ± 2.49	86.87 ± 2.93	89.09 ± 1.67	95.98 ± 1.10	**98.59 ± 0.87**
AA	90.22 ± 1.41	96.22 ± 2.67	91.46 ± 1.06	84.54 ± 3.76	82.10 ± 3.86	82.90 ± 3.78	93.03 ± 1.91	**97.71 ± 1.41**
Kappa	91.77 ± 1.40	96.67 ± 2.89	93.93 ± 0.92	88.63 ± 2.77	85.32 ± 3.29	87.84 ± 1.86	94.41 ± 3.80	**98.42 ± 0.97**

The proposed method SCL-Net achieves the highest value of the objective evaluation index. The OA, AA, and Kappa of the proposed method SCL-Net are increased by 16.65%, 13.86%, and 21.32% when compared with SSRN, respectively. The OA, AA, and Kappa of the proposed method SCL-Net are 3.21%, 4.22%, and 4.19% higher than the second-best method MTGAN, respectively. Besides, the classification accuracies of asphalt and gravel significantly improved from other methods. The classification maps of the proposed method and the comparison methods are shown in Fig. 8.

Tables VI and VII show the objective indicators on the Houston and KSC datasets. For each line, the bold fonts indicate the best performance. The proposed method obtains the highest value of OA (87.80%), while PCA-EPF acquires the lowest OA (74.35%) on the Houston dataset. For the KSC dataset, SCL-Net can acquire highest class accuracies' values for 11 out of the total 13 classes, which also proves the effectiveness of the proposed method. The classification maps on the Houston and KSC datasets are shown in Figs. 9 and 10. Although other methods consider the spatial–spectral characteristics of HSI, the misclassification phenomenon still occurs

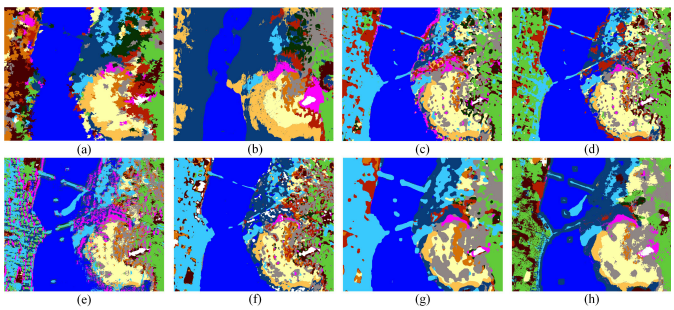

Fig. 10. Classification maps for the KSC dataset. (a) SBD. (b) PCA-EPF. (c) SPRN. (d) SSRN. (e) RSSAN. (f) A2S2K. (g) MTGAN. (h) SCL-Net.

resulting in the dot noise on the classification maps. This is because the spectral–spatial characteristics of those misclassified land-cover objects are very similar. It is worth noting that the proposed method SCL-Net can decrease the similarity of features between different samples by introducing supervised contrastive learning.

In addition, we also verify the classification performance under different training samples. Fig. 11 shows the overall

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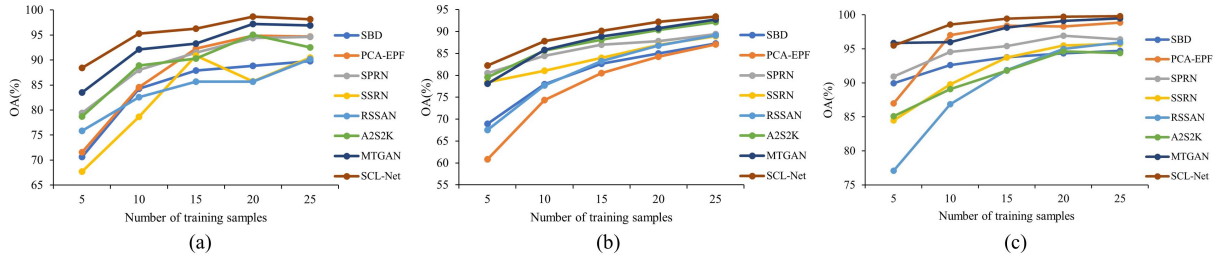

Fig. 11. Overall accuracies with respect to the number of training samples on three datasets. (a) PaviaU. (b) Houston. (c) KSC.

TABLE VIII
GPU INFERENCE TIME (S) OF THE PROPOSED METHOD AND THE COMPARISON METHODS

Method	SPRN [42]	SSRN [23]	RSSAN [43]	A2S2K [44]	MTGAN [27]	SCL-Net
PaviaU	25.90	22.88	9.97	50.43	5.29	7.10
Houston	88.69	100.96	33.50	227.59	20.69	25.63
KSC	38.63	59.77	17.36	86.72	10.11	14.04

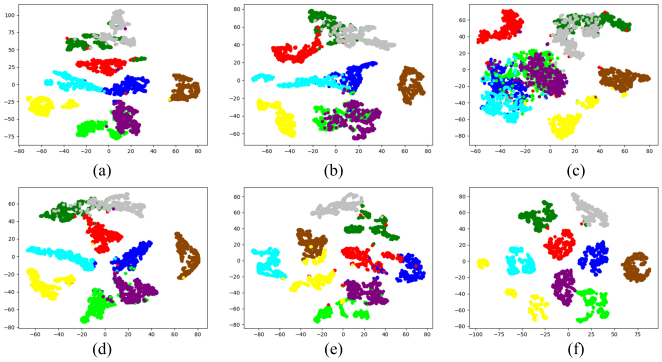

Fig. 12. t-SNE visualization results on the PaviaU dataset. 500 samples are randomly selected from the testing set for each object class. (a) SPRN. (b) SSRN. (c) RSSAN. (d) A2S2K. (e) MTGAN. (f) SCL-Net.

classification accuracy of the proposed method and comparison method under different number of training samples. The number of training samples for each category ranged from 5 to 25, with a gradual increase in step size of 5. In general, the proposed method consistently outperforms other methods on three datasets. The proposed method SCL-Net achieves the maximum overall accuracies of 98.68% when 20 training samples were selected for each category on the PaviaU dataset. For the Houston dataset, the overall accuracies of all the methods gradually improve with the increase in the number of training samples. Especially, it is worth noting that an obvious superiority of the proposed method is generated in the KSC dataset when the number of training samples is small. In overall, the experimental results show that the proposed method has better robustness and generalization ability, especially in the case of limited samples.

To more intuitively demonstrate the separability of the deep features generated by different DL-based methods, we use t-distributed stochastic neighbor embedding (t-SNE) [45], [46] for visualization of deep features. The t-SNE transforms the high-dimensional features into 2-D feature space for visualization, and the results are shown in Fig. 12. In all, 500 samples of each category from the testing set are randomly selected for visualization. The features corresponding to each category are displayed in different colors. It is obvious that the spectral–spatial features obtained by SCL-Net maintain better independence and separability in various categories than other comparison methods. This demonstrates that the proposed method could extract more discriminative features of interclass dispersion than the compared DL-based methods.

F. Time Consumption

The results of the GPU inference time of the proposed method and the compared DL-based methods are shown in Table VIII, which are obtained by calculating the test time on all the three datasets. All the experiments are implemented on a computing server equipped with an Intel Core i7-6850K CPU and an NVIDIA GeForce GTX 1080ti GPU. The GPU inference time varies with the spatial size of datasets. The larger the spatial size, the longer time it takes. Obviously, MTGAN takes the shortest time because of the lightweight model on the same dataset. Due to the use of 3-D convolution operation, the A2S2K, SPRN, and SSRN methods all cost more time. Although RSSAN uses a 2-D convolution operation, it takes more time due to the use of the residual module. The proposed SCL-Net and MTGAN methods perform competitive in time consumption, which show shorter GPU inference time than the other compared DL-based methods. Among the compared methods, SBD and PCA-EPF are non-DL-based methods and they are completed in MATLAB platform without GPU acceleration. Thus, for a fair comparison, the time of these two methods is not included here. Comprehensively, SCL-Net achieves the best performance in terms of all three quantitative indicators in less time.

IV. CONCLUSION

In this article, a novel end-to-end SCL-Net for HSI classification is proposed. The SCL-Net method mainly consists of two components: PCL and multilevel spectral–spatial information fusion (MLSIF). For the PCL course, supervised contrastive learning is introduced to extract high discriminative

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spectral features with intraclass compactness and interclass dispersion. The pairwise strategy enhances the learning ability for detailed information with limited samples. For the MLSIF course, the discriminative spectral features are merged with spatial features by the spectralwise multiplication operation. The multilevel decision fusion is designed to make full use of the complementary information. The experimental results show that the proposed method achieves better classification performance on three HSI datasets within an acceptable time.

In this article, since the pairwise labels are generated based on the center pixel, we designed a contrastive loss to measure the similarity between spectral pixels. Actually, it is also deserving to deeply explore an effective contrastive learning model in the spatial feature learning branch, which should be one future focused direction. Besides, except for the limited labeled samples, there are abundant unlabeled samples that are more easily to be obtained in practice. Therefore, how to mine more supervised information from both limited labeled samples and unlabeled samples and develop an end-to-end semisupervised contrastive learning based on the proposed SCL-Net algorithm will be another focus of future work.

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